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An assessment of PFAS contamination in wild boar meat and liver

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Abstract

In this study 266 wild boar samples (152 meat and 114 liver) from Germany were evaluated for contamination with four EU-regulated perfluoroalkyl substances (PFAS), namely perfluorooctane sulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), and perfluorohexane sulfonic acid (PFHxS). The arithmetic mean concentrations of the sum of these four PFAS were 5.18 µg/kg in meat and 448 µg/kg in liver and the corresponding median values were 1.24 µg/kg and 339 µg/kg, respectively. All liver samples (100%) and 12.5% of the meat samples exceeded the maximum levels set by the European Union (EU Regulation 2023/915), rendering them unmarketable. Finally, potential health risks associated with dietary exposure to wild boar meat and liver were assessed based on the tolerable weekly intake (TWI) of 4.4 ng/kg body weight for the sum of the four PFAS, assuming a human body weight of 70 kg.

Keywords Wild boar, PFAS, Maximum levels, Dietary exposure, Consumer risk, TWI

Introduction

Per- and polyfluoroalkyl substances (PFAS) are a large group of anthropogenic chemicals that do not occur naturally in the environment. Owing to their unique physicochemical properties they are widely used in numerous industrial products and processes, including non-stick coatings for cookware, textile impregnation, ski waxes, fire-fighting foams, food packaging, and many other applications [1]. A key characteristic of many PFAS compounds is their high thermal and chemical stability, as well as their grease-, water-, and dirt-repellent properties. Because of the strong carbon–fluorine bond PFAS are extremely persistent and cannot be naturally degraded in the environment either abiotically or biotically [2]. For this reason, they are often referred to as "forever chemicals." During the manufacture, use and disposal of PFAS-containing products these substances are released into the environment and are distributed globally via atmospheric and aquatic pathways making them ubiquitous. Plants absorb and accumulate PFAS from contaminated soil and water, while animals are primarily exposed through food intake and drinking water [3].

Among the large group of PFAS, the long-chain compounds perfluorooctane sulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA) and perfluorohexane sulfonic acid (PFHxS) contribute disproportionately to human exposure. PFAS are detectable in numerous foodstuffs with fish and meat often showing particularly high concentrations [4]. For most of the European population, food represents the primary source of exposure. Toxicological data is available mainly for individual PFAS, with PFOS and PFOA being the most extensively studied. These compounds have been associated with reduced antibody responses following vaccination, increased cholesterol levels, reduced birth weight, elevated alanine aminotransferase levels and other effects [5].

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In 2020 the European Food Safety Authority (EFSA) established a tolerable weekly intake (TWI) of 4.4 ng PFAS/kg body weight (bw) for the sum of four PFAS (PFOS, PFOA, PFNA, and PFHxS; ∑PFAS-4). These substances were selected because they contribute substantially to PFAS concentrations in human blood serum, exhibit relatively long biological half-lives, and were assumed to have similar toxicological potencies [5]. In 2023, EU Regulation 2023/915 introduced maximum levels for certain contaminants in food, including PFAS in game meat and game offal (Table 1). Foodstuffs exceeding one or more of these maximum levels may not be marketed in the European Union (EU) [6].

Wild animals accumulate xenobiotics in their tissues and can therefore serve as bioindicators of environmental contamination. The feeding behaviour of wild boar (*Sus scrofa*) includes burrowing in the ground and the consumption of a wide variety of foods including plants, animals, and fungi [7]. In combination with their omnivorous diet their relatively high trophic level makes wild boar particularly suitable as bioindicators. Wild boar are widely distributed across many regions of the world and are hunted for population control, allowing comparatively large numbers of samples to be obtained from hunters and forestry authorities for scientific studies [8]. Compared with meat from other animal species such as cattle, deer and poultry wild boar meat has been reported to be more highly contaminated with PFAS at statistically significant levels [9]. In addition, wild boar liver consistently shows higher PFAS concentrations than the livers of hare, chamois, red deer, roe deer, eider duck, beaver, and nutria [2].

The primary aim of this study was to assess PFAS contamination in wild boar meat and liver in relation to the maximum levels established by the European Union (EU Regulation 2023/915). For this purpose, concentrations of four PFAS (PFOS, PFOA, PFNA, and PFHxS) were compared with the corresponding EU maximum levels. To evaluate potential health risks, dietary exposure to wild boar meat and liver was assessed based on the TWI of 4.4 ng/kg body weight for ∑PFAS-4 defined by EFSA, assuming a human body weight of 70 kg. Both average consumers and heavy consumers were considered.

Materials and methods

Data basis

All data considered in this study, with the exception of the data from Hesse, were provided by Friends of the Earth Germany (BUND) [10]. Using the environmental information and transparency laws of the federal states, BUND requested and obtained data on PFAS concentrations in wild boar meat and liver from the responsible state authorities. Data from Hesse were obtained from the Hessian Ministry for the Environment, Climate Protection, Agriculture and Consumer Protection for evaluation under the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) framework by the European Chemicals Agency [11]. In all cases, at least four PFAS were analysed and reported in µg/kg fresh weight: PFOS, PFOA, PFNA, and PFHxS.

In total, data from 152 wild boar meat samples and 114 wild boar liver samples were evaluated for individual PFAS and for their sum (∑PFAS-4). Meat samples originated from Baden-Württemberg (n = 20), Bavaria (n = 22), Hesse (n = 49), Lower Saxony (n = 15), North Rhine-Westphalia (n = 5), Rhineland-Palatinate (n = 25), and Saxony (n = 16). Liver samples were provided by Bavaria (n = 7), Hesse (n = 70), Rhineland-Palatinate (n = 25), and Schleswig–Holstein (n = 12). The complete dataset is provided in the supplementary files (see Additional file 1).

Meat and liver samples were collected within the respective federal states and originated from various sources, including natural habitats, forestry authorities, and grocery stores. Exact information on the geographical origin of the animals was available for 95 samples. This information was not available for the remaining samples so that it cannot be excluded that some animals were shot in another federal state or even abroad rather

Table 1 Maximum levels of PFAS in game meat and offal per amended EU Regulation 2023/915

Perfluoroalkyl substances	Maximum level (µg/kg)				
	PFOS	PFOA	PFNA	PFHxS	Sum of PFOS, PFOA, PFNA and PFHxS
Meat of game animals, with the exception of bear meat	5.0	3.5	1.5	0.60	9.0
Offal of game animals, with the exception of bear offal	50	25	45	3.0	50

Maximum levels of PFOS, PFOA, PFNA, PFHxS and their sum in game meat and game offal; based on EU Regulation 2023/915 as amended on April 25, 2023

than in the federal state from which the respective sample was obtained.

BUND requested data on wild boar samples analyzed from 1 January 2023 onward. Most data originated from 2023; however, not all federal states reported the year of sampling. The dataset was analyzed in the form provided by the state authorities. Each laboratory in the participating federal states used its own method to determine PFAS. From 2020 to 2024, the European Union Reference Laboratory (EURL) conducted a laboratory comparison study for PFAS in fish fillets, whole eggs, honey, pork liver, fish meal, milk powder, mixed feed, beef, and infant formula. In 2024 fifty-eight laboratories participated in an investigation of beef samples with PFAS concentrations in the range of maximum levels. The results of the participating laboratories were highly comparable, and no evidence of systematic errors or deficiencies in the analytical methods was found [12].

Consumption of wild boar meat

Only limited data are available on the consumption of wild boar meat in Germany. A telephone survey conducted by the German Federal Institute for Risk Assessment (BfR) on infrequently consumed foods reported that 22% of respondents had never consumed wild boar meat, 43% had not consumed it within the previous 12 months, and the remaining 35% had consumed wild boar meat once or more than once per year [13].

According to the BfR, an average of one to two game meals containing 200 g of game meat per person per year are consumed [14]. It was therefore assumed that wild boar meat represented the only game meat consumed and that average consumers ingested 300 g of wild boar meat per year (1.5 meals × 200 g = 300 g/year). This corresponds to a weekly consumption of approximately 5.77 g of wild boar meat (300 g ÷ 52 weeks ≈ 5.77 g/week).

The BfR estimates that heavy consumers such as hunters and their households may consume up to 60 game meals of 200 g per person per year [14], corresponding to approximately 12,000 g per year and about 231 g per week (12,000 g ÷ 52 weeks ≈ 231 g/week). Thus, heavy consumers eat slightly more than one wild boar meal per week.

Consumption of wild boar liver

Data on the consumption of wild boar liver in Germany are also limited. According to the same BfR telephone survey, approximately 41.5% of respondents had never consumed game offal, 51% had not consumed it within the previous 12 months, and 7.5% had consumed game offal once per year or more frequently [13].

Assuming that average consumers ingest one liver portion of 125 g per year, this corresponds to approximately 2.40 g of wild boar liver per week (125 g ÷ 52 weeks ≈ 2.40 g/week). For heavy consumers, a larger annual portion of 250 g was assumed, corresponding to approximately 4.81 g per week (250 g ÷ 52 weeks ≈ 4.81 g/week). These consumption estimates are based on data reported by the BfR [15].

TWI calculations

The tolerable weekly intake (TWI) of 4.4 ng/kg body weight per week applies to the combined intake of PFOS, PFOA, PFNA, and PFHxS (∑PFAS-4). In this study, it was assumed that PFAS exposure originated exclusively from the consumption of wild boar meat or wild boar liver. Because these foods are consumed infrequently and only annual consumption data were available, annual intake values were converted to weekly consumption to enable comparison with the TWI. Potential cases exceeding the TWI were assessed separately for average and heavy consumers. The weekly PFAS intake was calculated for all samples by multiplying the PFAS concentration in wild boar meat or liver by the weekly consumption data and divided by the body weight of 70 kg. This results in the PFAS intake expressed as ng PFAS/kg bw × week.

$$I = (C \times M) / BW$$

I is the PFAS intake per bw per week (ng PFAS/kg bw × week), C is the PFAS concentration in wild boar meat or liver (ng PFAS/g wild boar meat or liver), M is the weekly consumption amount (g wild boar meat or liver/week), and BW is the body weight (70 kg).

Furthermore, the maximum amount of PFAS that a person weighing 70 kg may ingest per week when fully exhausting the TWI is calculated. Based on this value, the maximum tolerable weekly consumption of wild boar meat or wild boar liver is derived for the 25th percentile, median, and 75th percentile.

$$\begin{aligned} TWI_max &= TWI \times BW = 4.4\,ng\,PFAS/kg\,bw \\ &\quad \times week \times 70 = 308\,ng\,PFAS/week \end{aligned}$$

$$M_max = TWI_max / C$$

TWI_max is the maximum tolerable weekly PFAS intake (ng PFAS/week), TWI is the tolerable weekly intake (4.4 ng PFAS/kg bw × week), BW is the body weight (70 kg), M_max is the maximum tolerable weekly consumption of wild boar meat or liver (g wild boar meat or liver/week), and C is the PFAS

concentration in wild boar meat or liver (ng PFAS/g wild boar meat or liver).

Statistics

Data analysis programs

The data was analyzed using Microsoft Excel 2024 (Microsoft Corporation) and RStudio 2024 version 4.4.0 (Posit PBC). Tables were generated in Microsoft Excel and graphics were generated using RStudio. The package ggplot2 was used for creating figures, and the package xlsx was used for importing Excel files.

Detection and quantification limits

All values below the respective detection or quantification limits were set to zero using a lower-bound approach. Of the 266 individual data points this applied to 116 samples in which at least one PFAS concentration was below the detection or quantification limit. In these cases, only the affected PFAS value was set to zero, while quantified concentrations of the remaining PFAS remained unchanged.

Results and discussion

Presentation of the measured values in the context of current literature

A total of 266 wild boar samples were evaluated comprising 152 meat samples and 114 liver samples. With an arithmetic mean concentration of 448 µg/kg and a median of 339 µg/kg, wild boar liver was several times more highly contaminated than wild boar meat, which showed an arithmetic mean of 5.18 µg/kg and a median of 1.24 µg/kg for ∑PFAS-4 (Table 2).

PFAS were detected in all wild boar liver samples, whereas in 21 of the 152 meat samples concentrations remained below the limit of detection or quantification. Tavoloni et al. [18] likewise reported PFAS in all analyzed wild boar liver samples, while Arioli et al. [16] and Felder et al. [17] detected PFOS as the only PFAS in only a subset of wild boar meat samples, indicating consistently higher contamination levels in liver than in meat [16–18]. Overall, wild boar livers showed high PFOS concentrations as the liver acts as a primary target organ for PFOS and therefore accumulated this compound to a greater extent than muscle tissue [18]. Numata et al. [19] investigated the distribution of several PFAS including PFOS, PFOA, PFNA, and PFHxS, in domestic pigs after ingestion of contaminated feed. Most ingested PFAS were detected in blood plasma and edible tissues, particularly in blood and muscle, whereas PFOS showed a high affinity for liver tissue [19].

In most samples, PFOS accounted for the largest proportion of ∑PFAS-4 in both wild boar meat and liver. Based on median values, PFOS contributed approximately 81% of ∑PFAS-4 in meat and about 95% in liver. In wild boar meat, PFOA accounted for 13.4%, PFNA for 3.2%, and PFHxS for 2.5%, whereas in liver PFOA accounted for 1.9%, PFNA for 2.9%, and PFHxS for 0.4%. Median ∑PFAS-4 concentrations in wild boar liver were approximately 250 times higher than in meat; conversely, meat concentrations corresponded to about 0.4% of liver values. The PFOS-to-PFOA ratio was 6:1 in meat and 51:1 in liver, and PFOS concentrations in meat represented approximately 0.24% of those in liver. Neither PFNA nor PFHxS constituted the dominant PFAS in any sample. In 21 meat samples and in 5 liver samples,

Table 2 Mean, median and range of PFAS levels in wild boar meat and liver

	Mean [µg/kg]	Median [µg/kg]	25th quantile [µg/kg]	75th quantile [µg/kg]	Min [µg/kg]	Max [µg/kg]
Wild boar meat n = 152						
ΣPFAS-4	5.18	1.24	0.38	3.00	0	257
PFOS	1.55	0.73	0.20	2.00	0	17.9
PFOA	3.30	0.12	0	0.370	0	252.00
PFNA	0.22	0.03	0	0.105	0	4.60
PFHxS	0.12	0.02	0	0.079	0	2.30
Wild boar liver n = 114						
∑PFAS-4	448.18	339.37	222	526	98.2	3494.59
PFOS	366.95	299.75	201.88	476.05	75.0	1437.70
PFOA	61.75	5.85	3.60	8.40	0	3145.00
PFNA	17.07	9.25	6.53	12.70	0	273.00
PFHxS	2.41	1.19	0	3.08	0	25.80

Mean, median, minimum and maximum levels of PFOS, PFOA, PFNA, PFHxS and their sum (∑PFAS-4) in wild boar meat and wild boar liver. All values refer to fresh weight. Data in µg/kg; values below the respective detection or determination limit were set to zero; *n* number of samples

PFOA represented the largest fraction of total PFAS. For instance, the most contaminated meat sample, with a total PFAS concentration of 257 µg/kg, consisted of 98% PFOA.

PFOS was the dominant PFAS in 240 of the 266 samples analyzed, consistent with findings reported by Tavoloni et al. [18], Felder et al. [17], and Kowalczyk et al. [7] for wild boar liver and by Mikolajczyk et al. [9], Felder et al. [17], and Arioli et al. [16] for wild boar meat [7, 9, 16–18]. Cases in which other PFAS such as PFNA or PFOA occurred at higher concentrations than PFOS were generally associated with specific circumstances such as regional hotspots or particular environmental conditions. For example, elevated PFNA and PFOA concentrations reported by Schröder et al. [20] suggested a specific contamination source and/or differences in diet [20]. Data from Rupp et al. [8], as well as individual highly contaminated samples in the present study (see paragraph "PFAS hotspots"), indicated the presence of local hotspots [8]. Overall the results confirmed the high variability of PFAS concentrations in wild boar reported in the literature. Wild boar liver was particularly heavily contaminated, whereas contamination levels in wild boar meat showed greater variability and, in some cases, no PFAS were detected (Table 3).

EU maximum levels

Most wild boar meat samples contained PFAS concentrations well below the prescribed EU maximum levels. A total of 19 of the 152 samples exceeded at least one EU maximum level for an individual PFAS or for their sum. Consequently, 12.5% of the evaluated meat samples were not marketable in the EU. The EU maximum level was exceeded in 7 samples for PFOS, 8 samples for PFOA, 8 samples for PFNA, 6 samples for PFHxS, and in 11 samples for the sum of PFAS. The sample with the highest ∑PFAS-4 concentration (257 µg/kg) exceeded the EU maximum level of 9 µg/kg by a factor of more than 28 (Fig. 1).

All wild boar liver samples exceeded the EU maximum levels for PFOS and ∑PFAS-4; therefore, none of the evaluated liver samples were marketable in the EU. The liver sample with the lowest ∑PFAS-4 concentration (98.2 µg/kg) exceeded the EU maximum level of 50 µg/kg by a factor of approximately two, whereas the maximum concentration (3494 µg/kg) exceeded the EU limit by a factor of about 70. The median ∑PFAS-4 concentration of 339 µg/kg was approximately seven times higher than the prescribed EU maximum level. Only a limited number of samples exceeded the EU maximum levels for PFOA, PFNA, and PFHxS: six samples for PFNA, six for PFOA, and 29 for PFHxS (Fig. 2).

Individual liver-to-meat concentration ratios were calculated for paired wild boar samples from Rhineland-Palatinate (n = 25). Paired samples refer to liver and muscle tissues collected from the same individual animal, allowing a direct within-animal comparison of organ-specific PFAS accumulation. For all investigated PFAS, the observed ratios substantially exceeded the ratios implied by the respective EU maximum levels [6]. Median ratios ranged from 15.8 for PFHxS to 270 for PFOS, whereas the regulatory ratios derived from EU Regulation 2023/915 ranged from 5 to 30 (Table 4). This pronounced discrepancy reflects the strong preferential accumulation of PFAS in liver tissue over muscle and explains why liver samples frequently exceeded regulatory limits while most meat samples remained compliant. These findings underline that compliance with maximum levels in meat does not necessarily indicate low internal organ burdens in exposed animals.

Foodstuffs that exceed the EU maximum levels for PFAS must not be placed on the market. However, there is no general obligation to test all foods for PFAS prior to sale. Instead, the competent authorities of the EU Member States are required to perform random sampling within official food controls to monitor compliance with the established maximum levels [21]. As no mandatory pre-market control exists, wild boar meat and liver exceeding the specified maximum levels may nevertheless enter trade. Moreover, wild boar harvested by hunters for personal consumption are not subject to official controls, as these products do not enter the commercial market. Consequently, the TWI may be exceeded when wild boar meat or liver is consumed, particularly by heavy consumers such as hunters and their households.

In addition, although EU maximum levels are designed to protect against acute peak exposures, products with lower concentrations remain marketable. Long-term consumption of such products therefore contributes to continuous PFAS intake in addition to exposure from other foods and drinking water. Given the widespread use of PFAS and their numerous environmental entry pathways, a rapid and comprehensive phase-out of PFAS is necessary to prevent further contamination of the environment and food chains [3].

Tolerable weekly intake (TWI)

EFSA's TWI for PFAS indicates the amount of PFAS that a person can ingest weekly over a lifetime without expecting adverse health effects; this assessment must take into account that this consideration is primarily based on long-term intake and accumulation [4, 5]. Occasionally exceeding the TWI is therefore not necessarily associated with an immediate health risk. A single portion of

Table 3 Mean and median of PFAS levels in wild boar meat and liver as reported in various studies

References	Country	Tissue	Number of samples	Median (∑PFAS)	Median (PFOS)	Median (PFOA)	Median (PFNA)	Median (PFHxS)	Mean (∑PFAS)	Mean (PFOS)	Mean (PFOA)	Mean (PFNA)	Mean (PFHxS)
Present study (2026)	Germany	Meat	152	1.24	0.73	0.12	0.03	0.02	5.18	1.55	3.30	0.22	0.12
		Liver	114	339.37	299.75	5.85	9.25	1.19	448.18	366.95	61.75	17.07	2.41
Tavoloni et al. [18]	Italy	Muscle	26	n.a	n.a	n.a	n.a	n.a	0.949	n.a	n.a	n.a	n.a
		Liver	26	n.a	n.a	n.a	n.a	n.a	65.3	n.a	n.a	n.a	n.a
Felder et al. [17]	Germany	Muscle	57	n.a	7.29	n.d	n.d	n.d	n.a	8.1	n.d	n.d	n.d
		Liver	90	408.59	n.a	n.a	n.a	n.a	n.a	n.a	n.a	n.a	n.a
Mertens et al. [22]	Germany	Liver	82	90.2	81.4	1.78	4.36	0.26	n.a	n.a	n.a	n.a	n.a
		Muscle	73	1.03	0.63	0.12	0.09	0.07	n.a	n.a	n.a	n.a	n.a
Schröder et al. [20]	Czech Republic	Liver	29	n.a	170	75	230	n.d	n.a	n.a	n.a	n.a	n.a
Rupp et al. [8]	Germany. Area Paper sludges	Liver	9	432	n.a	n.a	n.a	n.a	n.a	n.a	n.a	n.a	n.a
	Germany. Area Industrial emissions	Liver	1	792 (only one sample)		n.a	n.a	n.a	n.a	n.a	n.a	n.a	n.a
	Germany. Area Background contamination	Liver	11	101	n.a	n.a	n.a	n.a	n.a	n.a	n.a	n.a	n.a
Mikolajczyk et al. [9]	Poland	Muscle	14	0.54	0.24	0.077	0.055	0.031	0.64	0.40	0.11	0.069	0.057
Arioli et al. [16]	Italy	Muscle	20	n.a	n.a	n.a	n.a	n.a	n.a	1.44	n.a	n.a	n.a
Kowalczyk et al. [7]	Germany	Liver	91	n.a	133	3.5	n.a	n.a	n.a	n.a	8.8	n.a	n.a
Mateus-Vargas et al. [25]	Germany. Federal State 1	Liver	44	n.a	276.8	5.9	8.0	0	n.a	n.a	n.a	n.a	n.a
	Germany. Federal State 2	Liver	128	n.a	194.5	7.3	10.1	0	n.a	n.a	n.a	n.a	n.a
	Germany. Federal State 3	Liver	45	n.a	83.4	2.0	3.0	0	n.a	n.a	n.a	n.a	n.a

n.a. not available (not reported), *n.d.* not detectable

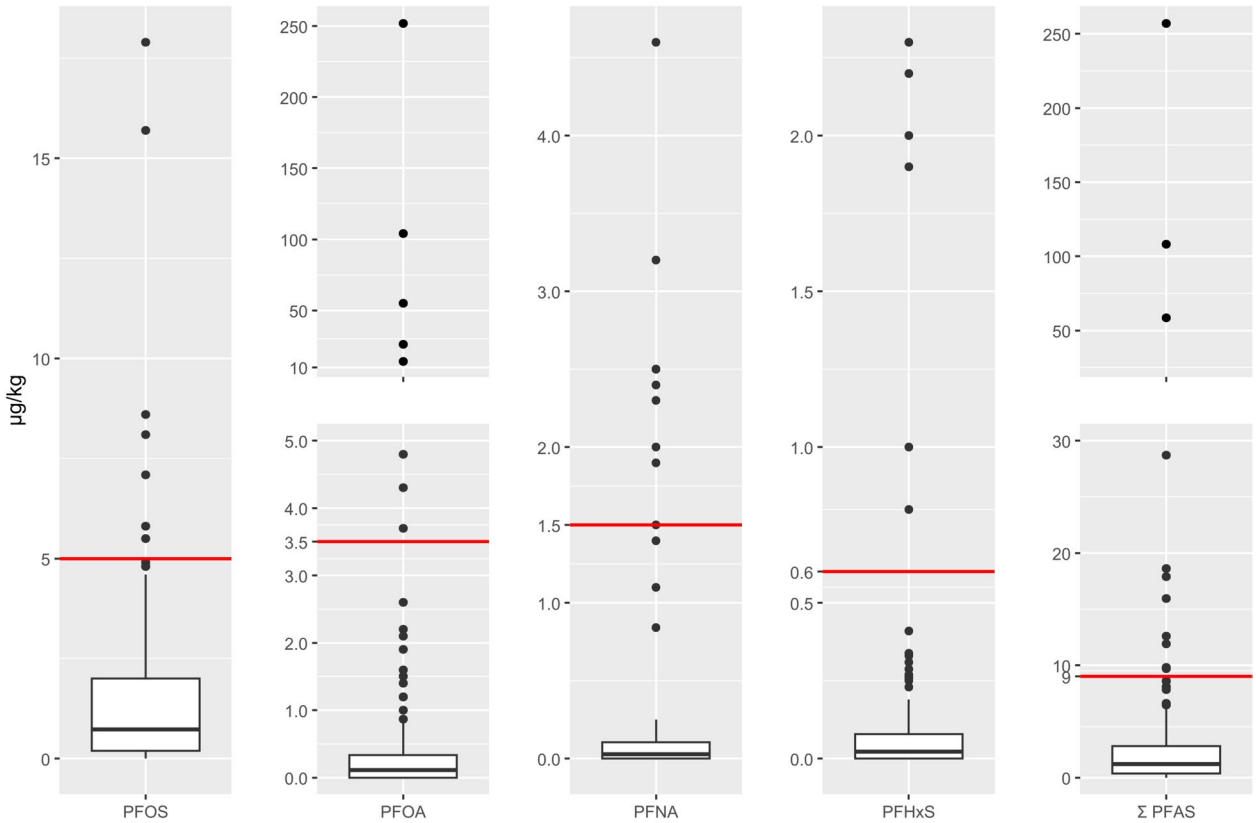


Fig. 1 Boxplots of PFAS levels in wild boar meat compared to EU Regulation 2023/915. Boxplots for the levels of PFOS, PFOA, PFNA, PFHxS, and their sum (∑PFAS-4) in wild boar meat and representation of the levels below and above the EU maximum levels for PFAS according to EU Regulation 2023/915. Data in µg/kg; number of samples was 152; the red line marks the respective maximum level according to the EU Regulation: PFOS: 5.0 µg/kg, PFOA: 3.5 µg/kg, PFNA: 1.5 µg/kg, PFHxS: 0.6 µg/kg, sum of the four PFAS: 9.0 µg/kg. Each data point represents one sample (data points may overlap). Samples above the red line exceed the permitted maximum levels and cannot be marketed.

contaminated wild boar meat or liver would only represent a health concern if such consumption occurred regularly over long periods, leading to sustained exceedance of the TWI. Consequently, assessment of average long-term intake based on consumption data is more relevant than isolated intake events.

As described in the materials and methods section, this study assumed that PFAS exposure originated exclusively from the consumption of wild boar meat or liver. This approach does not account for additional dietary sources such as other meats, meat products, fish, and fish products, which are known to contribute substantially to overall PFAS exposure [4]. Nevertheless, due to the comparatively high PFAS concentrations in wild boar tissues, consumption of wild boar meat or liver alone may already result in exceedance of the TWI, even without considering other exposure pathways.

Average consumers were assumed to consume approximately 300 g of wild boar meat per year (5.77 g/week), whereas heavy consumers consumed approximately 12,000 g per year (231 g/week). For wild boar liver average consumption was estimated at approximately 125 g per year (2.40 g/week), while heavy consumption was assumed to be 250 g per year (4.81 g/week). Table 5 summarizes consumption scenarios and the frequency of TWI exceedance for a body weight of 70 kg. For wild boar meat, PFAS intake of average consumers remained below the TWI in 98% of samples (149 of 152), exceeding the TWI in only 2% of samples (three samples). In contrast, consumption by heavy consumers resulted in TWI exceedance in 46% of samples (70 of 152). Contamination of wild boar liver was consistently higher: for average consumers, PFAS intake exceeded the TWI in 93% of samples (106 of 114), while for heavy consumers the TWI was exceeded in all evaluated cases (100%) (Table 5).

Consumption of the most highly contaminated wild boar meat sample (257 µg/kg) would have exceeded the TWI by a factor of approximately five for average consumers (5.77 g/week) and by a factor of 193 for heavy consumers (231 g/week). Consumption of the most

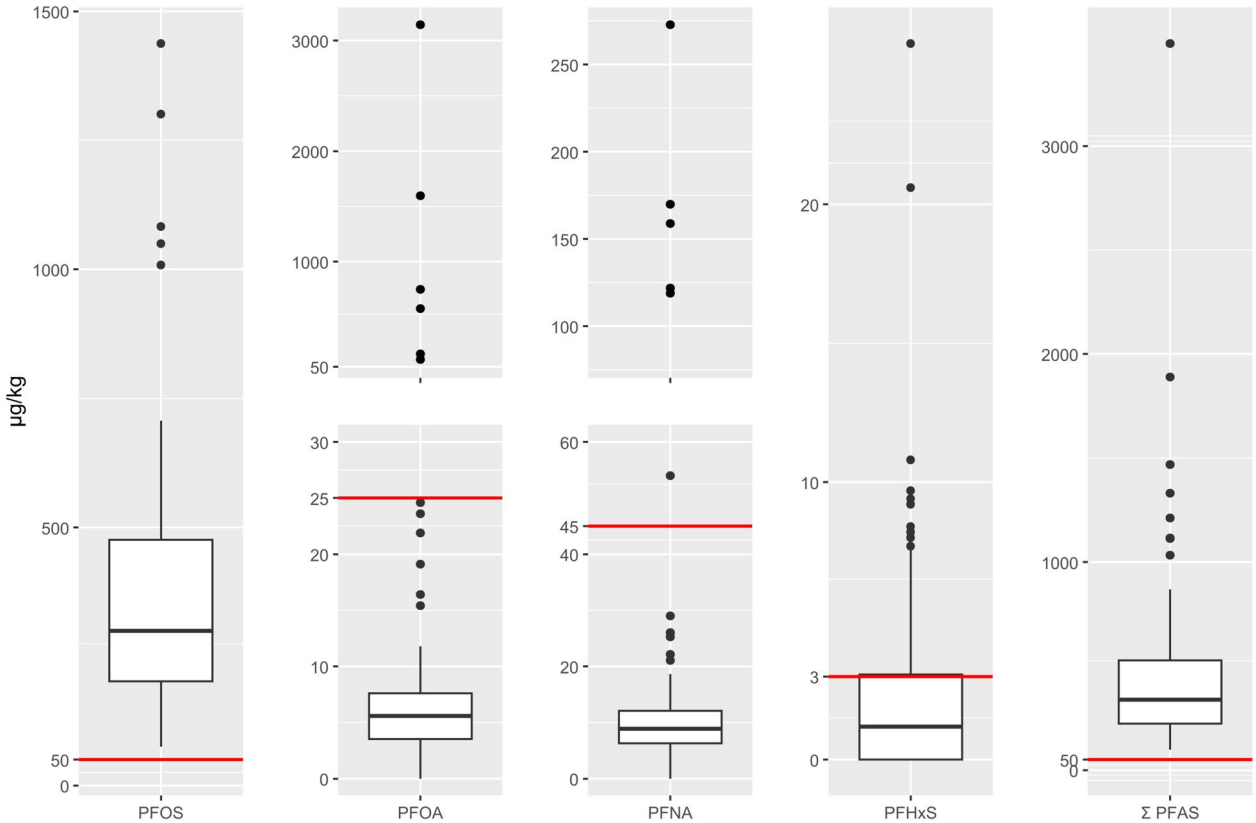


Fig. 2 Boxplots of PFAS levels in wild boar liver compared to EU Regulation 2023/915. Boxplots for the levels of PFOS, PFOA, PFNA, PFHxS, and their sum (∑PFAS-4) in wild boar liver and representation of the levels below and above the EU maximum levels according to EU Regulation 2023/915. Data in µg/kg; number of samples was 114; the red line marks the respective maximum level according to the EU Regulation: PFOS: 50 µg/kg, PFOA: 25 µg/kg, PFNA: 45 µg/kg, PFHxS: 3.0 µg/kg, sum of the four PFAS: 50 µg/kg. Each data point represents one sample (data points may overlap). Samples above the red line exceed the permitted maximum levels and cannot be marketed.

Table 4 Liver-to-meat PFAS ratios in wild boar compared with ratios implied by EU maximum levels

	EU-Regulation 2023/915	25th quantile	Median	75th quantile
ΣPFAS-4	50/9 ≈ 5.56	143	211	326
PFOS	50/5 = 10	182	270	453
PFOA	25/3.5 ≈ 7.14	19.7	29.5	42.3
PFNA	45/1.5 = 30	100	159	204
PFHxS	3.0/0.60 = 5	10.5	15.8	27.2

Liver-to-meat concentration ratios for ΣPFAS-4, PFOS, PFOA, PFNA and PFHxS in wild boar from Rhineland-Palatinate. Reported values represent the 25th percentile, median, and 75th percentile of individual paired samples (n = 25). The column "EU Regulation 2023/915" indicates the ratio implied by the respective maximum levels for liver and meat (maximum level in liver divided by maximum level in meat). Ratios are dimensionless

highly contaminated wild boar liver sample (3494 µg/kg) would have exceeded the TWI by a factor of 27 for average consumers (2.40 g/week) and by a factor of 55 for heavy consumers (4.81 g/week).

A total of 19 of the 152 wild boar meat samples exceeded the EU maximum levels and were therefore not marketable. The three samples that would have led to TWI exceedance for average consumers were among these 19 samples. For heavy consumers, consumption of 70 of the 152 samples would have resulted in TWI exceedance. While the EU maximum levels therefore protected average consumers from exceeding the TWI, this protection did not apply to heavy consumers. Accordingly, 51 samples exhibited PFAS concentrations below the EU maximum levels but would still have resulted in exceeding the TWI for heavy consumers. This indicates that the current maximum levels for wild boar meat are insufficient to protect heavy consumers from potentially harmful dietary PFAS exposure.

Contamination of wild boar liver was considerably higher than that of wild boar meat. All 114 liver samples exceeded the EU maximum levels and were therefore not marketable. For average consumers, consumption of 106 of these samples would have led to

Table 5 Consumption and TWI exceedances from wild boar meat and liver for average and heavy consumers

	Wild boar meat		Wild boar liver	
	Average consumers	Heavy consumers	Average consumers	Heavy consumers
Consumption per year [g]	300	12000	125	250
Consumption per week [g]	5.77	231	2.40	4.81
n	152	152	114	114
n below the TWI	149	82	8	0
n above the TWI	3	70	106	114
n above the TWI [%]	2	46	93	100

Amount of consumption and the number of times the TWI is exceeded when wild boar meat or wild boar liver is consumed by average or heavy consumers. *n* number of samples; TWI (∑PFAS-4) = 4.4 ng PFAS/kg body weight × week; the calculations are based on a body weight of 70 kg

TWI exceedance, whereas for heavy consumers all 114 samples would have exceeded the TWI. Because all wild boar liver samples were non-marketable irrespective of contamination level, the EU maximum levels in this case ensured protection against exceeding the TWI for both average and heavy consumers.

In the present study and in studies by Mikolajczyk et al. [9], Mertens et al. [22], Tavoloni et al. [18], and Rupp et al. [8], different consumption scenarios were considered in estimation of weekly PFAS intake from wild boar meat and liver. Tavoloni et al. [18], Rupp et al. [8], and the present study based their calculations on weekly consumption amounts, whereas Mikolajczyk et al. [9] and Mertens et al. [22] chose single-portion scenarios. Although the latter studies also suggested that the TWI may already be exceeded by consumption of a single portion of wild boar meat or liver, an assessment based on realistic weekly consumption is preferable, as it better reflects long-term exposure. Despite the uniform assumption of a body weight of 70 kg, the differing exposure scenarios limit direct comparability of results. Overall, the available studies indicate that consumption of wild boar meat only rarely results in exceedance of the TWI, whereas wild boar liver consistently showed frequent and often substantial TWI exceedances. [8, 9, 18, 22]. These findings were consistent with the results of the present study and highlighted the pronounced PFAS contamination of wild boar liver as reflected by different exposure scenarios. To enable a more robust and comprehensive health risk assessment of wild boar meat and liver consumption, improved data on actual consumption patterns in Germany would be highly desirable.

Table 6 summarizes the maximum amounts of wild boar meat and liver that a person weighing 70 kg could consume per week while fully exhausting the TWI through consumption of these products. Calculations were based on the 25th percentile, median, and 75th percentile PFAS concentrations. A typical portion size of 200 g was assumed for wild boar meat, whereas a portion size of 125 g was assumed for wild boar liver.

Table 6 Maximum tolerable weekly consumption of wild boar meat and liver under full exhaustion of the TWI

	25th quantile	Median	75th quantile
Wild boar meat [g/week]	811	248	103
Wild boar liver [g/week]	1.39	0.91	0.58

Amount of wild boar meat and liver (in grams per week) that can be consumed before fully exhausting the TWI, based on the 25th percentile, median, and 75th percentile of PFAS concentrations; TWI (∑PFAS-4) = 4.4 ng PFAS/kg body weight × week; the calculations are based on a body weight of 70 kg

Based on the median PFAS concentration, a maximum weekly consumption of 248 g of wild boar meat was acceptable, corresponding to slightly more than one portion. At the 75th percentile, the tolerable amount decreased to 103 g per week, equivalent to approximately half a portion. In contrast, for wild boar liver even very small quantities resulted in exhaustion of the TWI. At the median concentration, only 0.91 g per week could be consumed, and even at the 25th percentile the tolerable weekly amount was only 1.39 g. Consequently, a single annual portion of 125 g would already exceed the TWI, as 1.39 g/week × 52 weeks corresponds to approximately 72 g/year, which is well below one typical portion.

Felder et al. [17] calculated acceptable weekly intakes based on median PFAS concentrations of 7.25 µg/kg in wild boar meat and 408.59 µg/kg in wild boar liver, likewise assuming a body weight of 70 kg and a TWI of 308 ng PFAS per week. They derived maximum weekly consumptions of 24.25 g for wild boar meat and 0.75 g for wild boar liver. While the liver-related results of the present study were comparable to those reported by Felder et al. [17], the tolerable amount for wild boar meat was substantially higher in the present dataset, which can be explained by the lower median PFAS concentrations observed in meat samples [17].

Schuler et al. [23] likewise estimated tolerable weekly consumption amounts assuming a body weight of 70 kg. Using the 95th percentile concentration of 1.131 µg/kg in wild boar meat, the authors derived a maximum tolerable weekly consumption of approximately 272 g under full exhaustion of the TWI. Furthermore, they concluded that specific population groups, particularly heavy consumers who regularly consume wild boar meat and products, may easily exceed the TWI. This observation is fully consistent with the present results and further highlights the potential health risk associated with frequent consumption of wild boar products [23].

PFAS hotspots

A well-documented PFAS contamination exists in the district of Altötting (Bavaria), caused by fluoropolymer production at the Gendorf Chemical Park, where PFAS were historically released into the environment [24]. In this study, six paired wild boar meat and liver samples from this district were evaluated. Notably, in contrast to most other samples PFOA constituted the dominant PFAS in wild boar meat from Altötting. In liver samples, PFOA either represented the highest individual concentration or occurred at levels comparable to PFOS. The elevated PFOA contamination can be attributed to the Gendorf Chemical Park, where PFOA was produced between 1968 and 2003 and used as an emulsifier in fluoropolymer production until 2008 [24].

The two most highly contaminated wild boar liver samples, with ∑PFAS-4 concentrations of 3493 µg/kg and 1889 µg/kg, both originated from this district. Likewise, the four most contaminated wild boar meat samples (257 µg/kg, 108 µg/kg, 58.6 µg/kg, and 28.7 µg/kg ∑PFAS-4) were found in animals from Altötting. From a statistical perspective, these extreme values contributed substantially to the divergence between arithmetic means and median concentrations. Of the 152 meat samples, only three would have resulted in the TWI being exceeded for average consumers, and all three originated from Altötting. Based on the available data, consumption of wild boar meat from non-hotspot areas did not appear to result in TWI exceedance for average consumers.

In the present study, the median PFOS-to-PFOA ratio was 6:1 in wild boar meat and 51:1 in wild boar liver. Kowalczyk et al. [7] reported a ratio of 20.5:1 in wild boar liver [7], and a comparable background ratio of 22:1 was observed by Rupp et al. [8]. In contrast, samples from Altötting in the present study exhibited an inverse ratio, similar to observations at a fluoropolymer production site reported by Rupp et al. [8]. Specifically, the PFOS-to-PFOA ratio in Altötting samples was 1:37 in meat and 1:3 in liver, whereas Rupp et al. [8] reported a ratio of approximately 1:10 in liver. This inversion was most likely attributable to historical PFOA emissions associated with fluoropolymer production at the Gendorf Chemical Park.

Mateus-Vargas et al. [25] demonstrated that land-use patterns influence PFAS concentrations and may promote the formation of regional contamination hotspots [25]. It is therefore possible that additional samples in the present dataset originated from other hotspots; however, this could not be further assessed due to limited information on sample origin and the absence of identifiable contamination sources.

Consumption recommendations

In this study PFAS exposure was assumed to originate exclusively from the consumption of wild boar meat or liver. However, other relevant dietary sources, such as fish and fish products as well as meat and meat products, also contribute substantially to total PFAS intake and were not considered in this assessment [4]. The fact that the TWI could be exceeded solely through consumption of wild boar meat or liver highlights the high level of contamination of these foods. Felder et al. [17], for example, concluded that PFAS contamination of wild boar liver was sufficiently high to render it unsafe for consumption [17]. Accordingly, the German Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection advises against the consumption of wild boar liver [26]. Similar recommendations have been issued at the state level, for example by the Hessian Ministry for the Environment, Climate Protection, Agriculture and Consumer Protection [27]. In addition, wild boar liver and meat are used in processed products, such as sausages, meat-based dressings, and pâté. Although such products were not investigated in the present study, they are likely to contribute substantially to dietary PFAS exposure.

All calculations in this study were based on a reference body weight of 70 kg. Individuals with lower body weight, particularly children, may therefore experience higher PFAS intake per kilogram of body weight when consuming comparable amounts of contaminated food, resulting in a more rapid exceedance of the TWI. In their study, Mikolajczyk et al. [9] examined wild boar meat contaminated with PFAS and concluded that the consumption of wild boar meat is a relevant source of PFAS exposure, especially for children [9].

Limitations

The samples evaluated in this study originated from different sources, and the exact geographical origin was not available for all samples. This was partly due to

incomplete documentation and the fact that some samples were obtained from grocery stores without explicit origin labelling. Consequently, precise attribution of samples to specific regions was not always possible and it cannot be excluded that some samples originated outside Germany or that certain contamination hotspots remained undetected. Nevertheless, assessment of PFAS contamination in wild boar meat and liver available on the German market remains highly relevant.

The data for this study was collected by various laboratories. It should be noted that these laboratories used analytical methods with different detection and quantification limits. In the laboratories of some German federal states, the detection and quantification limits are higher than in others, so that certain PFAS were not detectable in the samples tested, although they possibly would have been detected in another laboratory. However, it can at least be assumed that the laboratories of the German federal states have adhered to the requirements of Implementing Regulation (EU) 2022/1428 [28]. Interpretation of the results was further limited by incomplete information on animal sex, age, and precise origin. Similar limitations were reported by Mateus-Vargas et al. [25], who noted that such constraints do not affect the central conclusions regarding compliance with maximum levels and exceedance of the TWI. To minimize these limitations in future studies, Mateus-Vargas et al. [25] recommended the implementation of standardized guidelines for sampling and data collection in PFAS monitoring of wild animals, enabling more robust regional comparisons and improved scientific evaluation [25].

The TWI and EU maximum levels currently consider only PFOS, PFOA, PFNA, and PFHxS, whereas other PFAS are not included. Moretti et al. [29] reported that the combined concentrations of chloropolyfluoropolyether carboxylates (ClPFPECAs) and their dechlorinated congeners (HPFPECAs) represented the fourth largest contribution to total PFAS concentrations in wild boar liver, following PFOS, PFOA, and perfluorotridecanoic acid (PFTrDA) [29]. Tavoloni et al. [18] likewise observed that although PFOS dominated PFAS concentrations in wild boar liver, PFTrDA exceeded PFOS concentrations in wild boar meat. The authors therefore questioned whether PFOS, PFOA, PFNA, and PFHxS alone are sufficient indicators for comprehensive assessment of PFAS contamination in food and suggested that inclusion of additional indicator compounds may allow more accurate exposure evaluation [18]. In the meantime, the EFSA opinion [5] on which this paper is based should be further developed since the EFSA assessment which assigns a relative potency factor (RPF) of 1 to each of the only four PFAS considered is becoming outdated. A proposal for this can be found, for example, in Bil et al. [30]. These authors determined the relative potencies of 16 PFAS for liver toxicity and suggested using RPF to estimate the combined effects of PFAS mixtures. Further suggestions are cited in reference [3].

Conclusion

This study assessed the occurrence of PFOS, PFOA, PFNA, PFHxS and their sum in wild boar meat and wild boar liver in relation to the maximum levels established by the European Union and evaluated the potential health risks associated with dietary exposure. The results clearly demonstrate that PFAS concentrations in wild boar liver are substantially higher than those in wild boar meat. All analyzed liver samples exceeded the applicable EU maximum levels and were therefore not compliant with food safety regulations, whereas 12.5% of the meat samples were classified as non-marketable.

With respect to the TWI, the findings indicate that the current EU maximum levels are generally sufficient to protect average consumers. However, the data further suggests that heavy consumers may exceed the TWI even when consuming products that comply with regulatory limits. Although wild boar meat and liver are not commonly consumed by the general population, specific subgroups, particularly hunters and their families, may experience a relevant contribution to overall PFAS exposure from these foods. According to the German Hunting Association, there were 460,771 hunting license holders in Germany in 2024 [31]. Existing advisories against the consumption of wild boar liver therefore appear justified, and additional regulatory measures may be warranted to further minimize potential health risks.

Based on the evaluated samples, consumption of wild boar liver is strongly discouraged, as even very small weekly amounts may lead to a lifetime exceedance of the TWI. Occasional consumption of wild boar meat by average consumers is considered toxicologically acceptable; however, it may contribute to cumulative dietary PFAS exposure. Meat originating from known contamination hotspots should be avoided, and heavy consumers are advised to limit their intake in order to prevent elevated exposure.

Beyond their relevance as a food source, wild boar also serve as bioindicators of the widespread occurrence of PFAS in the environment and food chains, thereby underscoring the broader environmental significance of these findings.

Abbreviations

BfR	Bundesinstitut für Risikobewertung (German Federal Institute for Risk Assessment)
BUND	Bund für Umwelt und Naturschutz Deutschland (Friends of the Earth Germany)
EFSA	European Food Safety Authority
EU	European Union

bw	Body weight (kg)
n	Number of samples
PFAS	Per- and polyfluoroalkyl substances [µg/kg or ng/g (fresh weight)]
PFOS	Perfluorooctane sulfonic acid [µg/kg or ng/g (fresh weight)]
PFOA	Perfluorooctanoic acid [µg/kg or ng/g (fresh weight)]
PFNA	Perfluorononanoic acid [µg/kg or ng/g (fresh weight)]
PFHxS	Perfluorohexane sulfonic acid [µg/kg or ng/g (fresh weight)]
RPF	Relative potency factor
TWI	Tolerable weekly intake (ng/kg bw/week)
∑PFAS-4	Sum of four PFAS (PFOS, PFOA, PFNA, and PFHxS) [µg/kg or ng/g (fresh weight)]

Supplementary Information

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Additional file 1. levels of PFOS, PFOA, PFNA, PFHxS, and their sum in wild boar meat and liver from various federal states in Germany

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Marleen Brossmann conducted the research, analyzed the data, and wrote the manuscript. Ingo Valentin had the fundamental idea for the topics. Hubertus Brunn provided supervision and conceptual guidance. All authors contributed to revisions and approved the final manuscript.

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Data availability
The data are provided in the supplementary files [see Additional file 1].

Declarations

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Not applicable.

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